Choosing the channels

Which filters, and what does each one contribute?

Notebook companion: `Unit-10-channels.ipynb`

1 The question

The instrument carries five filters. This unit says why those five, what each one contributes, and what happens to the measurement when one of them cannot be supplied.

Three separate things are being chosen at once, and they are easy to confuse. A filter determines the range of tangent altitudes at which its channel produces a usable measurement. It also determines how strongly that channel distinguishes one species from another. And the set as a whole determines how many species can be separated at a given altitude from the measurements alone, rather than from the assumed shape of the profile. The first is coverage, the second is contrast, and the third has no standard name in this project; we call it separation and define it in section 10c.

2 Physical picture

Two different technologies appear in the channel set, and they fail in different ways.

The three extreme-ultraviolet channels are free-standing metal foils, a few hundred nanometres thick, with no substrate. Zirconium passes 6 to 20 nm, aluminium 17 to 80 nm, indium 70 to 100 nm. A foil this thin has two hardware problems that a coated filter does not. Light can reach the detector through a pinhole without passing through the metal at all, and a pinhole does not discriminate by wavelength. And the metal reacts with its environment over time, so the transmission changes across a mission.

The two far-ultraviolet channels are dielectric multilayer coatings on a magnesium fluoride or calcium fluoride substrate, covering 135 to 155 nm and 155 to 175 nm. These are a different manufacturing process from a different supplier. Their bands are placed to exclude the Lyman-α line at 121.6 nm and the atomic oxygen resonance triplet at 130.4 nm, both of which are bright enough to dominate a band that admitted them.

What a channel measures follows from Unit 4. Above 115 nm the cross sections of atomic oxygen and molecular nitrogen are zero on our grid, so the two far-ultraviolet channels respond to molecular oxygen alone. Below 115 nm all three species absorb, and the ratio of their cross sections is what allows the species to be told apart.

Where a channel is usable follows from Unit 6. Its optical depth must lie inside the window 0.20 to 4.00. Below the lower wall the atmosphere is too transparent for the absorption to be measured against the calibration error; above the upper wall too little light is transmitted. Because the effective cross sections differ from channel to channel, each channel reaches that window at a different altitude, and the set covers a wider altitude range than any single filter could.

3 Derivation

Coverage and contrast both follow from the effective cross section. For a channel with response $r(\lambda)$ and solar flux $\Phi(\lambda)$, the optically thin band average is

$$\sigma_s^{\mathrm{eff}} \;=\; \frac{\int \Phi(\lambda)\, r(\lambda)\, \sigma_s(\lambda)\, \mathrm{d}\lambda}{\int \Phi(\lambda)\, r(\lambda)\, \mathrm{d}\lambda},$$

which `occ.sigma_eff` computes. The channel reaches the middle of its window where $\sum_s \sigma_s^{\mathrm{eff}} N_s \approx 1$, and the ratio $\sigma_{\mathrm{N_2}}^{\mathrm{eff}}/\sigma_{\mathrm{O}}^{\mathrm{eff}}$ is the factor by which the channel responds more strongly to nitrogen than to oxygen. Unit 4 records the accuracy limits of this approximation: it is correct to 0.3 per cent at an optical depth of 0.017 and 22 per cent high at 1.43, so it is used for comparing channels and not for converting a measurement into a column.

Separation needs a different quantity, because it is a property of the set rather than of one filter. Let $J_{cs} = \partial\tau_c/\partial \ln N_s$ for the channels whose optical depth is inside the window at one tangent altitude. Weighting each channel by its measurement error gives the Fisher matrix

$$F \;=\; J^{\mathsf{T}} W J, \qquad W = \mathrm{diag}\big(1/\sigma_c^2\big)\,,$$

and the marginal one-sigma error on $\ln N_s$ is $\sqrt{(F^{-1})_{ss}}$. When fewer channels are in window than there are absorbing species, F is singular. The species whose axes have a component along a null direction are then not determined at all: any value the retrieval returns for them at that altitude came from the prior or from the assumed profile shape, not from the measurement. `occ.py` uses the same eigen-decomposition guard in Unit 6, and the notebook reproduces it here.

Two cautions about this quantity. It is evaluated at one tangent altitude, whereas the parametric fit uses every altitude at once and imposes a shape, so the fit returns numbers where the pointwise calculation returns none. And it is a bound on precision, not a statement about bias.

4 Worked numbers

10a. What each channel measures

Channel	O	N_2	O_2	N_2/O	flux
Zr 6–20 nm	3.14	4.10	6.31	1.31	2.29×10^{9}
Al 17–80 nm	8.79	17.42	19.57	1.98	2.11×10^{10}
In 70–100 nm	2.26	12.60	16.05	5.57	2.10×10^{10}
FUV 135–155 nm	0	0	11.83	—	1.09×10^{11}
FUV 155–175 nm	0	0	1.76	—	7.13×10^{11}

Effective cross sections in megabarns ($10^{-18}\,\mathrm{cm}^2$) and admitted solar photon flux in $\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$. The two far-ultraviolet ratios are undefined because both denominators are zero.

Indium responds to nitrogen 5.57 times more strongly than to oxygen. That ratio is 4.3 times zirconium's and 2.8 times aluminium's. That single ratio is the origin of most of what follows. Zirconium and aluminium sit at 1.31 and 1.98, close enough to each other that an increase in the oxygen column and an increase in the nitrogen column produce nearly proportional changes in both

of those channels. Two measurements that respond in nearly the same way to two different changes cannot separate them.

The admitted flux varies by a factor of 300 across the set, but no channel is short of photons. The photon budget in Unit 9 shows why: even the zirconium channel, the faintest of the five, delivers far more signal than the measurement requires.

10b. Where each channel is usable

Channel	quiet	severe storm
Zr 6–20 nm	168–320 km	175–410 km
Al 17–80 nm	228–370 km	268–500 km
In 70–100 nm	168–310 km	218–458 km
FUV 135–155 nm	142–192 km	168–290 km
FUV 155–175 nm	115–155 km	118–200 km
at least one	115–370 km	118–500 km

Tangent altitudes at which the channel's optical depth lies inside 0.20 to 4.00. Both walls of that window are instrument specifications rather than properties of the atmosphere, and Unit 8 found that the two had been derived from different tolerances; settling them is an open item, and the figures here use the window currently in the code.

The five bands overlap in sequence, and the lowest and highest are set by different filters in the two atmospheric states. Two consequences.

The far-ultraviolet channels are the only ones usable below 168 km in quiet conditions, so the lower part of the covered range depends entirely on them. Under a severe storm every band moves up, because the atmosphere expands and a given slant column is found at a greater altitude. The shift is not uniform: the indium band moves 50 km and the aluminium band 40 km, while the 155–175 nm band moves only 3 km, so the lowest usable altitude changes from 115 to 118 km.

The covered range stops at 115 km whichever filters are fitted. Every channel is opaque below that altitude, and Unit 8 showed that widening the optical-depth window does not change it: a saturated channel is replaced by a weaker one only while a weaker one exists, and 155 to 175 nm is the weakest absorption in the set. Reaching the 90 km floor of the benchmark would require a band in which molecular oxygen absorbs more weakly still, and our cross-section and flux tables stop at 175.0 nm, so no such channel can be evaluated with the present data.

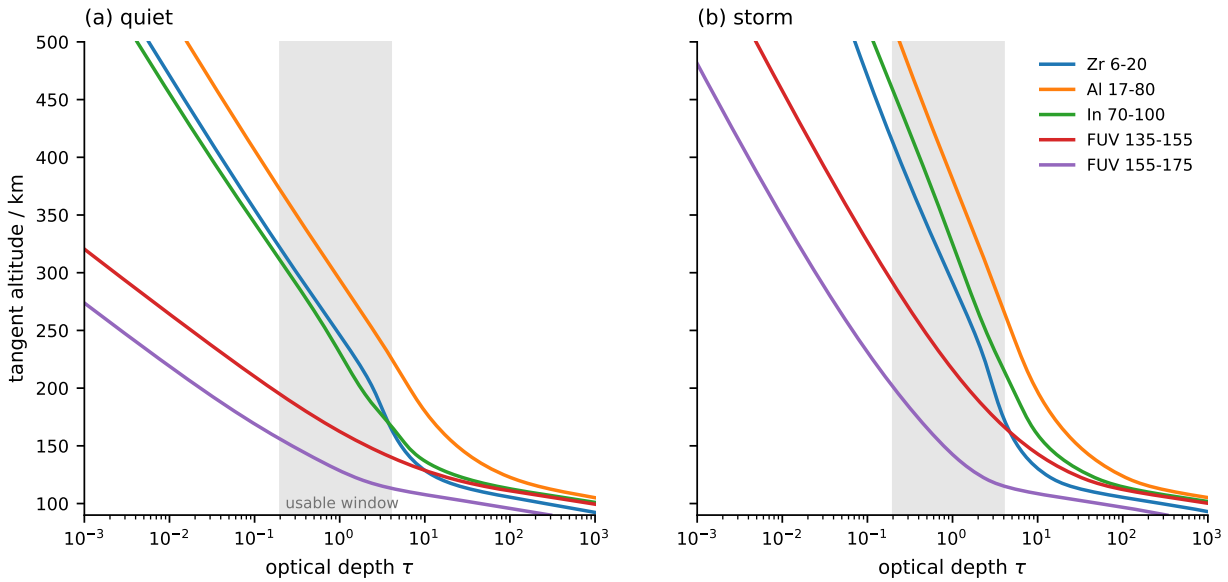


Figure 1: Optical depth against tangent altitude for each channel, with the usable window shaded. A channel is usable where its curve crosses the shaded band. The sequence of crossings is what produces the altitude coverage in the table above, and the whole set of curves shifts upward under storm conditions. Generated by section B of the notebook.

10c. How many species the measurement separates

Channel set	ch	300 km O	300 km N_2	300 km O_2	400 km O	400 km N_2	400 km O_2
Zr + Al	2	—	—	—	—	—	—
Zr + Al + In	3	29.6	13.7	29.9	24.1	30.2	193.9
Zr + Al + 2 FUV	2	—	—	—	—	—	—
all five	3	29.6	13.7	29.9	24.1	30.2	193.9

Marginal one-sigma error on each slant column, in per cent, from the channels in window at that altitude alone, in a severe storm. "ch" is the number of channels in window, which is the same at both altitudes shown. A dash means the species is not determined by the measurements at that altitude, because the Fisher matrix is singular in that direction. Lower down the pattern differs: at 200 km the two configurations containing far-ultraviolet channels determine molecular oxygen to 0.9 per cent and neither of the other two species at all.

This is the central finding of the unit. **Two channels cannot separate three absorbing species at a single altitude, and no arrangement of two filters changes that.** Where only zirconium and aluminium are in window, the pointwise problem has two measurements and three unknowns, and every species axis has a component along the unconstrained direction.

Counting altitudes over the 140 to 450 km range on a 5 km grid gives the extent of the difference. Zirconium and aluminium alone separate all three species *nowhere*, in either atmospheric state. Adding indium separates all three over 230 to 310 km in quiet conditions, a span of 85 km, and over 270 to 410 km in a severe storm, a span of 145 km. The full five-channel set extends that to 170–310 km quiet and 220–410 km storm.

The far-ultraviolet pair added without indium is not a substitute. It separates all three species over 270 to 290 km in a storm, a span of 25 km, and nowhere at all in quiet conditions. The reason is in section 10a. Above 115 nm the oxygen and nitrogen cross sections are zero, so wherever the extreme-ultraviolet channels are the only ones responding to those two species, the set is a two-channel set however many filters are fitted. What the far-ultraviolet channels add at lower altitudes is molecular oxygen alone: at 200 km in a storm they determine it to 0.9 per cent while neither of the other two species is determined at all.

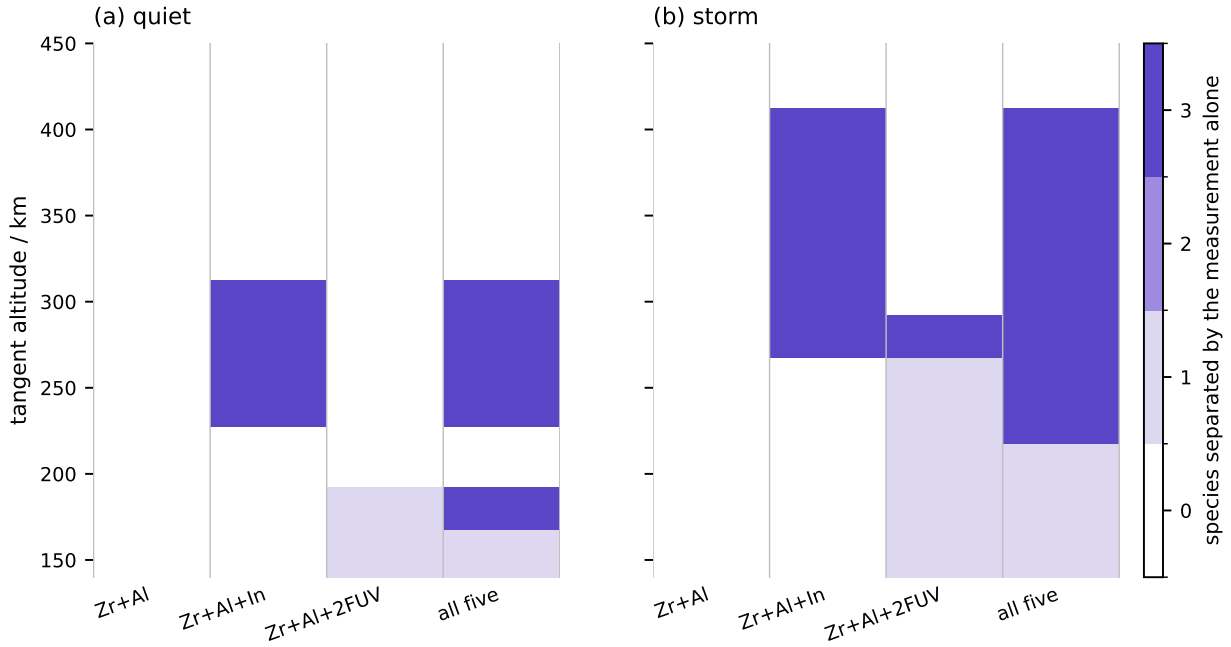


Figure 2: The number of species determined by the measurements alone at each tangent altitude, for the four channel sets. A value of three means the three slant columns are separately determined there; a value of zero means the measurements at that altitude constrain no species individually and any number the retrieval reports comes from its assumed profile shape. Generated by section C of the notebook.

This does not mean the retrieval fails where the count is below three. The parametric fit uses every altitude at once and imposes diffusive equilibrium on a Bates temperature profile, which supplies what the pointwise measurement does not. That is the correct reading, and it is also the uncomfortable one: the fewer altitudes at which the measurement separates the species by itself, the more of the retrieved answer rests on the shape assumption, and Unit 8 measured what that assumption costs when a storm violates it. A seven per cent reduction in the oxygen column over a 50 km layer moves the retrieved density outside the ten per cent benchmark.

10d. What each channel set delivers

		quiet		severe storm	
Channel set	**pts**	O	N_2	O	N_2
Zr + Al	43	$+0.93 \pm 4.38$	-2.40 ± 8.51	$+7.83 \pm 12.42$	-1.17 ± 2.28
Zr + Al + In	67	$+0.62 \pm 1.13$	$+0.42 \pm 1.11$	-0.74 ± 2.43	$+0.89 \pm 1.26$
Zr + Al + 2 FUV	62	$+14.62 \pm 3.27$	-20.61 ± 6.97	$+13.89 \pm 10.58$	-4.28 ± 2.53
all five	86	$+4.05 \pm 0.98$	-0.25 ± 1.29	$+1.50 \pm 2.02$	-1.41 ± 1.10

Bias and spread of the retrieved slant column at 200 km, in per cent, O_2 held, 30 draws, 10 km grid, Abel penalty 1.25, `occ.noise_sigma` carrying shot noise and flat-field drift. Settings identical to Unit 8 section 8b, so these reproduce that table rather than approximating it. "pts" is the number of channel–altitude points in window in the storm case.

Exospheric temperature, on a truth value of 838 K quiet and 1955 K storm: zirconium and aluminium give -0.6 ± 12.3 K and -77.9 ± 28.9 K; adding indium gives $+6.0 \pm 6.1$ K and -49.8 ± 13.3 K; the far-ultraviolet fallback gives $+5.1 \pm 9.2$ K and -114.6 ± 25.0 K.

Three readings, and the third is the one that changes a project decision.

Indium reduces the storm oxygen spread from 12.42 to 2.43 per cent and the bias from +7.83 to −0.74. Both halves of the ten per cent benchmark are met with indium in the set and neither is met without it.

The far-ultraviolet pair, added without indium, makes the oxygen bias worse rather than better: +13.89 per cent in a storm against +7.83 for the two extreme-ultraviolet channels alone, and +14.62 against +0.93 in quiet conditions. Section 10c gives the mechanism. Those two channels respond to molecular oxygen alone, and O_2 is held at its prior in this configuration, so their residuals depend only on a fixed quantity and on the two shape parameters. They therefore constrain the temperature profile tightly while adding no information about oxygen or nitrogen, and the fitted profile shape is then pulled toward satisfying them. Unit 8 measured the check on this explanation: releasing O_2 reduces the storm bias from +13.89 to +6.24 per cent.

The five-channel set is not the best set by this measure. In quiet conditions it returns $+4.05 \pm 0.98$ per cent against $+0.62 \pm 1.13$ for three channels, and its exospheric temperature error is +35.9 K against +6.0 K, for the same reason. More filters is not the same as more information.

WHY NOT Zr+Al+2 FUV as the fallback

The 5 September decision records indium as an enhancement and the two far-ultraviolet channels as an equivalent fallback if Luxel cannot supply it. On the corrected noise model that is not supportable. In a severe storm the fallback returns oxygen at $+13.89 \pm 10.58$ *per cent against* -0.74 ± 2.43 *with indium, and in quiet conditions* $+14.62 \pm 3.27$ *against* $+0.62 \pm 1.13$*. It fails both halves of the benchmark in both atmospheric states, and it is worse than the two extreme-ultraviolet channels it was meant to improve on. The mechanism is understood, not mysterious: the far-ultraviolet channels constrain the temperature profile through a species held at its prior. This does not reopen the channel set, because indium is already in the baseline. It changes what happens if the vendor answer is negative, and that is a decision recorded as open.*

10e. What the filters are made of, and how they fail

The dominant hardware term for the metal foils is out-of-band leakage. `occ.OOB_LEAK` is defined against the in-band signal, so it is multiplied by the ratio of out-of-band to in-band light. That ratio is large. Counting in detector electrons rather than photons, because an extreme-ultraviolet photon liberates several electron–hole pairs in silicon and a visible photon liberates one, out-of-band light arrives with 6.5×10^{6} times the signal of the zirconium channel and 1.7×10^{6} times that of aluminium.

pinhole	Zr	Al	In
0.1 µm	7.9×10^{-5}	2.1×10^{-5}	5.2×10^{-5}
0.2 µm	5.0×10^{-3}	1.3×10^{-3}	3.3×10^{-3}
0.5 µm	1.2	0.32	0.81
1.0 µm	20	5.4	13

Leak from one circular pinhole in one 25 µm × 10 mm slit, relative to the in-band signal, with Bethe's fourth-power suppression applied for holes small against 500 nm. The error budget assumes 10^{-3}.

A single hole larger than about 0.2 µm anywhere in one slit removes that channel from use. Holes below 0.15 µm are harmless, because a hole small against the wavelength transmits far less than its area suggests. The threshold is sharp, and it falls in the size range where real pinholes occur.

The film itself is a second term. The only published figure available to us is a visible rejection near 10^{-9} for a 1720 Å aluminium film, from Powell et al. (1990). Applied to our channels that gives a leak of 1.7×10^{-3} in aluminium and 6.5×10^{-3} in zirconium, against the 10^{-3} the error budget assumes. Both figures are questions in the Luxel RFI rather than conclusions; a stacked second film for visible blocking is the standard answer and is already why indium is carried at about one per cent net throughput.

The far-ultraviolet channels do not share this failure mode, because a coating on a substrate has no pinholes in the same sense. They have their own: coating uniformity across the slit, and the placement of the band edges, which must exclude two bright emission lines.

5 What the computation shows

Two channels cannot separate three species at one altitude. The Fisher matrix is singular in that case, so the retrieved value for every species at that altitude rests on the assumed profile shape rather than on the measurements. Zirconium and aluminium alone achieve three-species separation nowhere between 140 and 450 km. Adding indium achieves it over 145 km of that range in a storm and 85 km in quiet conditions; adding the far-ultraviolet pair instead achieves it over 25 km in a storm and nowhere in quiet conditions.

The ratio of the nitrogen to the oxygen effective cross section is what separates indium from the other two metal filters: 5.57, against 1.31 for zirconium and 1.98 for aluminium. Zirconium and aluminium have ratios close to each other, so a change in the oxygen column and a change in the nitrogen column produce nearly proportional changes in both of those two channels.

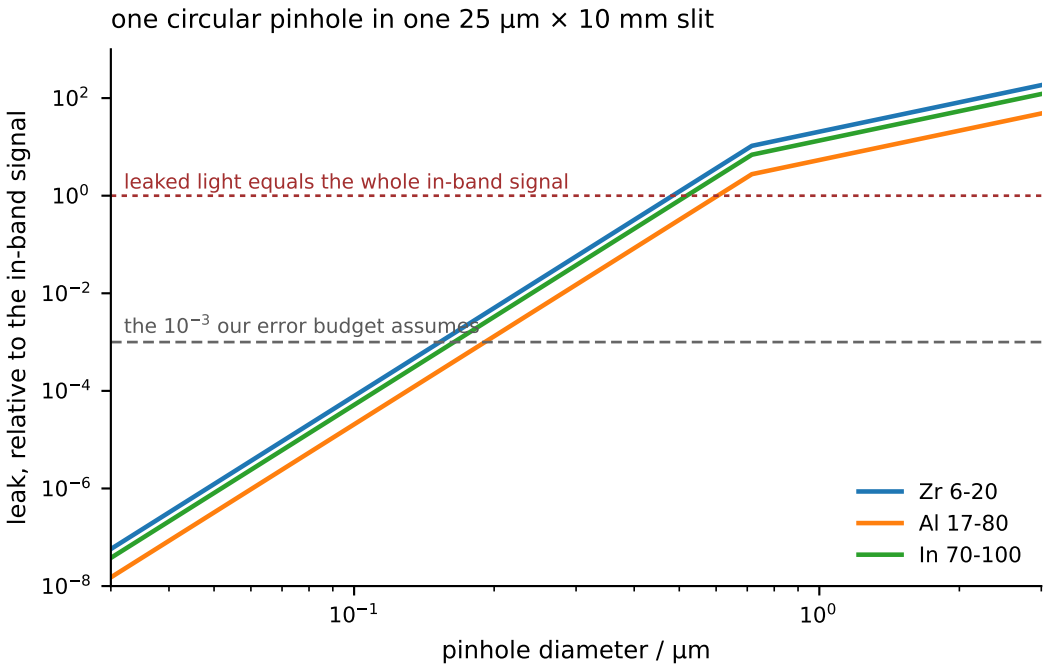


Figure 3: Leak from one circular pinhole against its diameter, for the three metal channels, with the assumed 10^{-3} marked. The steep rise between 0.1 and 0.5 µm is Bethe's suppression giving way as the hole approaches the wavelength of the light leaking through it. Generated by section E of the notebook.

Adding filters is not the same as adding information. The five-channel set is worse than the three-channel set in quiet conditions, at +4.05 per cent against +0.62, because two of its channels constrain the temperature profile through a species that is held at its prior.

The altitude range stops at 115 km for a reason no filter in the set can change. Every channel is opaque below it, and the weakest absorption available is the 155 to 175 nm band. Reaching lower requires a band we cannot presently evaluate, because the cross-section tables stop at 175.0 nm.

One pinhole can remove a channel. Above about 0.2 µm in diameter, a single hole in one slit admits more out-of-band light than the channel's entire in-band signal at 0.5 µm.

6 What we decided

The baseline channel set is zirconium, aluminium, indium and the two far-ultraviolet bands, which is the widest set that fits on one slit plate and was fixed on 5 September. This unit does not reopen it.

The fallback named on 5 September is withdrawn as an equivalent. Zirconium, aluminium and the two far-ultraviolet channels do not meet the benchmark in either atmospheric state, and they are worse than zirconium and aluminium alone. What replaces indium if the vendor answer is negative is an open decision, recorded in `STATUS.md` and due before the 3 October sensor decision. The candidates are a two-channel configuration with a restricted altitude claim, a different third extreme-ultraviolet filter, and releasing O_2 in the fit so that the far-ultraviolet channels contribute usefully.

Every reported channel-set comparison carries the atmospheric state, the noise model, the tangent-altitude grid and whether O_2 was held. Section 10d is sixteen numbers for four configurations of the same instrument, and only the labels distinguish them.

The Luxel RFI asks for pinhole density above 0.2 µm, per lot, because that is the threshold section 10e locates. It also asks whether stacking two films reduces the through-hole density as the product of the two.

7 What we do not claim

Not that the filter bands are the ones we will fly. Every band in this unit is an idealized rectangular passband. Real transmission curves have sloped edges and structure inside the band, and Unit 4 showed that a one-nanometre shift of an edge near 91 nm changes a species-separation ratio by an order of magnitude. This is the first of the four standing caveats on the whole project, and it applies to this unit more directly than to any other.

Not that the separation count predicts the retrieval's accuracy. It is a pointwise bound on precision from the measurements at one altitude. The parametric fit pools altitudes and imposes a shape, so it returns values where the pointwise calculation returns none. The count says how much of the answer is measured, not how wrong it is.

Not that the pinhole numbers are a specification. The out-of-band flux is an order-of-magnitude estimate rather than an integral over the detector response; the detector is taken as one electron per out-of-band photon; and Bethe's law is the ideal thin-screen limit, so a real hole in a film of finite thickness transmits less. The threshold near 0.2 µm and the ordering across channels are what the calculation supports.

Not that the throughput and ageing figures are ours. About one per cent net throughput for a stacked indium filter and about 30 per cent transmission loss in ten months are taken from the literature. They are in the Luxel RFI as questions.

Not that the optical-depth window is settled. Unit 8 found its two walls derived from different tolerances, and the upper one assumed an out-of-band leak ten times worse than the one we require. Every altitude band in section 10b moves if that window moves.

8 Exercises

E10.1 Section 10d shows the five-channel set performing worse than the three-channel set in quiet conditions. Run the same comparison with O_2 released rather than held, and report whether the ordering reverses. Then state, in one sentence, what a reader should conclude about reporting a channel-set comparison without saying which species were held.

E10.2 The separation count in section 10c uses the window currently in `occ.py`. Recompute it with the upper wall at 3.05, the value Unit 8's rule gives for the out-of-band rejection we require, and report how many kilometres of three-species separation are gained or lost in each atmospheric state. Say whether that changes which channel set you would choose.

E10.3 Indium distinguishes nitrogen from oxygen with a cross-section ratio of 5.57. Search the cross-section tables for a rectangular band between 6 and 100 nm that gives a higher ratio while admitting at least as much solar flux as the zirconium channel. If you find one, say what would

have to be true of a filter for it to be manufacturable; if you do not, say what that implies about how much of the channel set is set by physics rather than by choice.

Every quantitative result in this unit is provisional in the four ways that apply to the whole project: filter bands are idealized top-hats rather than measured transmission curves; the truth atmosphere is NRLMSIS 2.1; helium is omitted above 400 km; and no horizontal structure along the ray is modelled. Analysis code is in `channel-plan/`.